



2 Real Analysis 2023

Tutorial 10

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Compact sets in $\mathbb{R}$

Problem 39 Decide whether the following sets are compact. Give a justification.

- (a) $\mathbb{N}$,
- (b) $\mathbb{R}$,
- (c) $A = \{x \in [0, 10] : x \neq 5\}$,
- (d) $B = \{x \in \mathbb{R} : 1 \leq |x| \leq 2\}$,
- (e) $C = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$,
- (f) $D = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\}$.

Limits of functions

Problem 40 Prove by definition of the limit:

- (a) If $f(x) = c$ with some $c \in \mathbb{R}$ for all $x \in A$, and if a is a limit point of A , then $\lim_{x \rightarrow a} f(x) = c$.
- (b) If $f(x) = x$ for all $x \in A$, and if a is a limit point of A , then $\lim_{x \rightarrow a} f(x) = a$.

Problem 41 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $a, L \in \mathbb{R}$. Show that $\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{h \rightarrow 0} f(a + h) = L$.

Problem 42 Show that $\lim_{x \rightarrow 0} \cos\left(\frac{1}{x}\right)$ does not exist, but $\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$.

Problem 43 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with $\lim_{x \rightarrow 3} f(x) = 2$. Prove by the definition of the limit that

$$(a) \lim_{x \rightarrow 3} (f(x) + 1)^2 = 9, \quad (b) \lim_{x \rightarrow 3} \frac{2f(x) - 3}{4f(x) - 9} = -1.$$